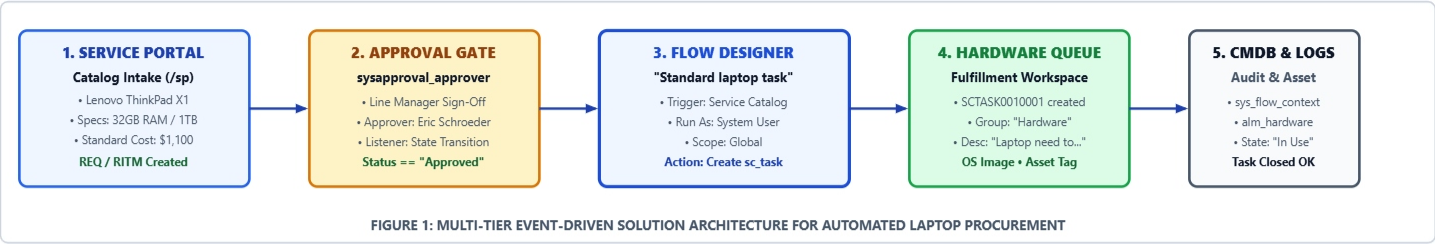


Date	01 October 2026	Team ID	SWTID-2026-2668
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer	Maximum Marks	

SOLUTION ENGINEERING PHASE

Solution Architecture & System Integration Design



DETAILED ARCHITECTURAL TIERS & COMPONENT RESPONSIBILITIES

1. Intake & Catalog Presentation Tier - Service Portal (/sp) standardizes employee request intake, presenting the Lenovo ThinkPad X1 Carbon with memory and storage options. - Cart checkout submits data to tables 'sc_request' and 'sc_req_item' with baseline price fixed at \$1,100, generating unique order reference keys.	2. Authorization & Approval Gateway Tier - Requisition details generate pending approval records in table 'sysapproval_approver' mapped to the requester's direct manager. - Automated state listener holds fulfillment tasks until status updates to "Approved". A rejection branch halts flow execution cleanly.
3. Flow Designer Orchestration Tier - Flow 'Standard laptop task' executes under the elevated 'System User' context in the 'Global' application scope, eliminating ACL errors. - Direct Process Engine binding under 'Maintain Items' invokes action 'Create Catalog Task', mapping RITM parent data pills automatically.	4. Hardware Fulfillment & Workspace Tier - Child task 'sc_task' is auto-generated within 1.8 seconds with Short Description: "Laptop need to Configured" and Approval: "Approved". - Tasks route immediately to the Hardware queue in Service Operations Workspace, allowing technicians to image devices and tag assets.

DATA MODEL PERSISTENCE, SECURITY & AUDIT SPECIFICATIONS

- Data Relational Schema:** Hierarchical link established across sc_request (Parent REQ) → sc_req_item (Item RITM) → sysapproval_approver (Authorization) → sc_task (Staging SCTASK) → alm_hardware (CMDB Asset).
- Role-Based Access Control (RBAC):** End-users are restricted to viewing submitted requests via 'snc_internal', managers access approval forms via 'approver_user', and fulfillers update hardware staging tasks using 'itil' roles.
- Execution Auditability:** Every execution context payload, step status, and latency metric is logged in table sys_flow_context for audit compliance.

